

ALWAYS FREE COMPUTE, WITHOUT THE CLOUD FOG

From empty tenancy to useful Linux server

A field guide to Oracle Cloud Infrastructure (OCI): what the pieces mean, which free VM to choose, how to connect, how traffic reaches it, and how to avoid surprise costs or preventable lockouts.

BEST FREE TARGET

A1

Ampere Arm flexible VM.
Allocate from the monthly
Always Free compute pool.

RECOMMENDED OS

Ubuntu

Familiar tooling and a broad
package ecosystem. Verify
every container image
supports arm64.

ACCESS MODEL

SSH

Use a key pair, not a server
password. The private key
stays on your own device.

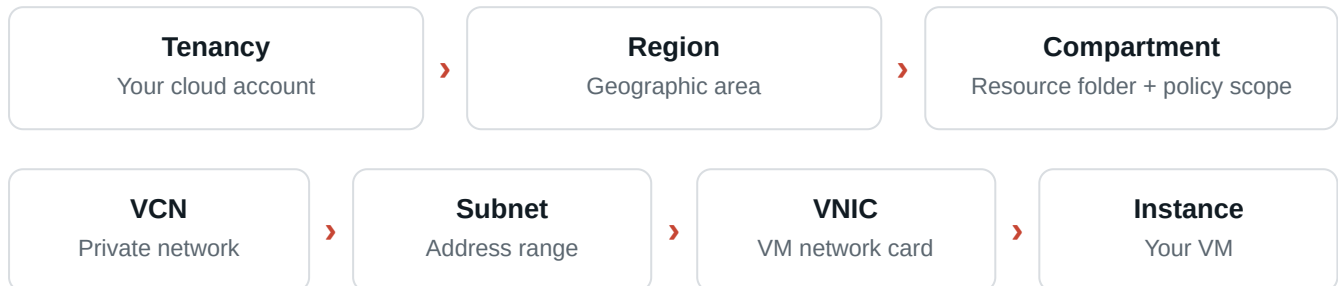
The free tier is a limit, not a magic shield. Confirm the resource is marked **Always Free eligible**, use only the home region for free compute, and create budget alerts before experimenting.

What this guide gives you

1. The OCI mental model	2
2. Your full Always Free VM inventory	3
3. Your VM compared with Sock's sandbox	4
4. A safe first-VM recipe	5
5. SSH and first-hour setup	6
6. Networking and firewall logic	7
7. Daily operations and troubleshooting	8
8. Cost, backup, and security guardrails	9
9. Glossary and official sources	10

OCI is a set of nested boundaries

Think from the outside inward. Most console confusion comes from viewing the right service in the wrong region or compartment.



Identity and organization

Tenancy: the top-level account and billing boundary.

Identity domain: users, sign-in, groups, and authentication.

Compartment: a logical container used for permissions, quotas, budgets, and finding resources.

OCID: Oracle Cloud Identifier, a long unique ID attached to each resource.

Compute and storage

Instance: a virtual machine.

Shape: CPU architecture, OCPU count, and memory profile.

Image: operating system template, such as Ubuntu or Oracle Linux.

Boot volume: persistent disk holding the operating system and files.

Location

Home region: the account's anchor region. Always Free compute must be created here.

Availability domain (AD): an isolated data-center group within a region.

Fault domain: a hardware grouping inside an AD.

Networking

VCN: your software-defined private network.

Subnet: a slice of the VCN address space.

Internet gateway: a route target that lets a public subnet reach the internet.

NSG: Network Security Group; firewall rules for selected VNICs.

Fast orientation: check the region selector, then the compartment selector, before assuming a resource is missing.

You can run up to four free VMs

As of September 2026, an Always Free tenancy can hold **up to four compute instances**: as many as **two Ampere A1 Flex VMs** sharing one Arm resource pool, plus **two AMD E2.1.Micro VMs**.

A1 FLEX ALLOWANCE

2 VMs

They share 2 OCPUs and 12 GB RAM across the tenancy.

E2 MICRO ALLOWANCE

2 VMs

Each has 1 AMD OCPU, 2 vCPUs, and 1 GB RAM.

MAXIMUM COUNT

4 VMs

Only if the shared 200 GB disk pool can hold all boot volumes.

The four slots are not identical

Free allowance	CPU	RAM	Best use
Up to 2 × A1 Flex	Ampere Altra Arm; 2 OCPUs total shared	12 GB total shared	Main app server, Docker, agents, Node/Python workloads
2 × E2.1.Micro	AMD x86; 1 OCPU = 2 vCPUs each via hyperthreading	1 GB each	Light bot, status page, proxy, tiny sidecar

How the A1 pool splits: one A1 can use the full **2 OCPUs / 12 GB**, as your planned VM does. If you want two A1 VMs, they must divide that same pool, for example **1 OCPU / 6 GB each**. The pool was reduced from 4 OCPUs / 24 GB in June 2026.

Storage determines how many fit

All boot and block volumes share **200 GB total**. OCI boot volumes have a 47 GB minimum. Four minimum-size boot volumes need about **188 GB**. Your 100 GB boot volume leaves 100 GB, enough for two more 47 GB boot volumes, so the practical maximum with the current disk choice is **three VMs total**, not four.

Other free infrastructure

- 10 TB outbound bandwidth each month
- Public IPv4 is free while attached to a running instance
- 200 GB combined boot/block storage

Two important caveats

- Arm capacity can be scarce: retry after `Out of host capacity`, or try another available region.
- Oracle may reclaim Always Free instances it classifies as idle.

Your Oracle VM is the bigger Linux box

nelson-vm-01: provisioning on 24 Sep 2026

This compares the Linux sandbox that runs Sock's tools with the Oracle VM you configured. It does **not** compare against the Meta GPU infrastructure that runs the Muse model itself.

	Sock's sandbox	nelson-vm-01
CPU	2 shared vCPUs	2 dedicated OCPUs
Processor	AMD EPYC 9D25	Ampere Altra
Architecture	x86_64	Arm64 / aarch64
Memory	7.7 GB RAM	12 GB RAM
Disk	7.5 GB	100 GB boot volume
Operating system	Ubuntu 24.04	Canonical Ubuntu 24.04 Minimal
CPU scheduling	Shared slices of a larger host	Full physical Arm cores; no hyperthreading
Role	Temporary tool-running workspace	Your persistent API-and-agent server

Your exact deployment plan

Name: nelson-vm-01

Shape: VM.Standard.A1.Flex

Region: ca-montreal-1

Image: Canonical Ubuntu 24.04 Minimal aarch64

Compute: 2 OCPUs / 12 GB RAM

Storage: 100 GB boot volume

Address: reserved public IPv4

Ingress: TCP 22, 80, and 443

Where your VM wins: 55% more RAM, over 13 times the disk, and dedicated physical CPU cores. It is built to stay online and host your own services.

Where the sandbox differs: x86_64 has broader compatibility with old or x86-only binaries. On the A1 VM, prefer packages and container images that publish linux/arm64 builds.

The “brain” is separate from the hardware. The VM runs the installed agent, command-line interfaces, orchestration, storage, and networking. API keys let those tools call remote large language models; the model inference usually runs at the API provider, not on the two Arm cores.

Build one simple, recoverable server

1 Name and placement

Open **Compute** › **Instances** › **Create instance**. Use `nelson-vm-01`. Confirm the compartment and the Montreal home region (`ca-montreal-1`).

2 Choose the image

Select **Canonical Ubuntu 24.04 Minimal aarch64** for A1. If you choose Oracle Linux, remember the SSH user changes from `ubuntu` to `opc`.

3 Choose the shape

Select `VM.Standard.A1.Flex`, then set **2 OCPUs / 12 GB RAM**. Stop if the page does not show the shape as Always Free eligible.

4 Create the network path

Use a VCN with a **public subnet**, internet gateway, and route to `0.0.0.0/0`. Assign a public IPv4 address if you need direct SSH from home.

5 Use an SSH public key

Upload an existing `.pub` key or let OCI generate a key pair. Download the private key once and store it securely. Oracle does not retain a recoverable copy of a console-generated private key.

6 Keep storage modest

Use the selected **100 GB** boot volume. It leaves 100 GB in the Always Free storage pool; every additional instance needs at least 47 GB.

7 Review before launch

Confirm the Always Free label, architecture, OCPU/RAM, boot size, public IP choice, and estimated cost. Then create the instance and wait for **Running**.

Open only what you use

- TCP 22: SSH, ideally from your own public IP
- TCP 80: HTTP, if serving a public website
- TCP 443: HTTPS, if serving a public website

Do not do this

- Do not paste a private SSH key into `cloud-init`.
- Do not enable password SSH for convenience.
- Do not expose databases or admin panels to `0.0.0.0/0`.

Arm compatibility check: before deploying a container, look for a multi-architecture image or explicit `linux/arm64` support. Run `uname -m`; A1 should report `aarch64`.

Your first ten minutes on the VM

1. Lock down the key locally

```
chmod 600 ~/keys/oracle-vm.key
```

2. Connect to the public IP

```
# Ubuntu image
ssh -i ~/keys/oracle-vm.key ubuntu@203.0.113.10

# Oracle Linux image
ssh -i ~/keys/oracle-vm.key opc@203.0.113.10
```

Replace the example address with the instance's real public IP. On first connection, verify the host-key fingerprint through a trusted channel before accepting it.

3. Identify and update the system

```
uname -m           # A1 normally reports aarch64
cat /etc/os-release
sudo apt update && sudo apt upgrade -y  # Ubuntu
# Oracle Linux: sudo dnf upgrade -y
```

4. Check resources

```
nproc              # logical processors visible to Linux
free -h            # memory
lsblk               # disks and mount points
df -h /             # root filesystem usage
ip -brief address   # interfaces and addresses
```

5. Create a memorable local SSH alias

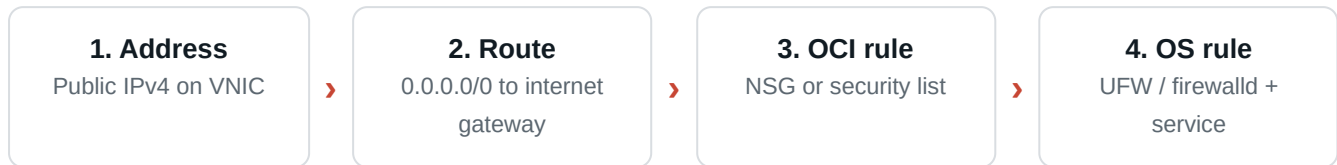
```
# ~/.ssh/config on your own computer
Host oracle-free
  HostName 203.0.113.10
  User ubuntu
  IdentityFile ~/keys/oracle-vm.key
  IdentitiesOnly yes

# Then connect with:
ssh oracle-free
```

Never lose the private key. Keep an encrypted backup. Anyone who gets it can attempt to access the server; Oracle's documentation says it cannot recover a console-generated private key for you.

File transfer: use `scp local-file oracle-free:~/` to upload, or `scp -r oracle-free:~/folder ./` to download.

A public IP alone does not open a port



Inbound traffic works only when every relevant layer permits it and the application is actually listening.

Layer	Question	Where to check
Public address	Does the primary VNIC have a public IP?	Instance > Networking > Attached VNICs
Route table	Does the subnet route internet traffic to an internet gateway?	VCN > Subnets > Route table
Cloud firewall	Does an ingress rule allow the protocol, port, and source?	NSG or subnet security list
Host firewall	Does Linux permit the port?	<code>sudo ufw status</code> or <code>firewalld</code>
Application	Is something listening on the correct interface?	<code>sudo ss -ltnup</code>

Security list

Rules apply to all VNICs in the associated subnet. Simple, but broad.

Network Security Group

Rules apply only to selected VNICs. Oracle recommends NSGs for separating application security from subnet design.

Source CIDR: choose the narrowest safe value

Source	Meaning	Use
<code>your.public.ip/32</code>	One IPv4 address	Best for SSH when your address is stable
<code>0.0.0.0/0</code>	Every IPv4 address	Normal for public HTTP/HTTPS; risky for admin ports
VCN CIDR	Private network only	Internal services between OCI resources

Two-firewall trap: adding an OCI ingress rule does not update Ubuntu's UFW or Oracle Linux's `firewalld`. Check both. Also bind public services to `0.0.0.0` or the VM's private address, not only `127.0.0.1`.

Run it like a small Linux server

System health

```
uptime
free -h
df -h
du -sh /* | sort -h
top
sudo ss -lntup
```

Services and logs

```
systemctl --failed
sudo systemctl status nginx
sudo systemctl restart nginx
sudo journalctl -u nginx -n 100
sudo journalctl -u nginx -f
```

Updates and reboot

```
sudo apt update
sudo apt upgrade -y
sudo reboot

# Oracle Linux
sudo dnf upgrade -y
```

Docker basics

```
docker ps
docker compose ps
docker compose logs -f
docker compose pull
docker compose up -d
docker system df
```

When SSH does not connect, check in this order

1. **Lifecycle:** the instance is **Running**, not Stopped, Terminated, or Provisioning.
2. **Address:** use the current public IP shown on the instance details page.
3. **Network:** public subnet, internet-gateway route, and TCP 22 ingress are present.
4. **Source:** if port 22 is limited to a `/32`, your current home public IP still matches it.
5. **Username:** `ubuntu` for Ubuntu; `opc` for Oracle Linux and Red Hat-compatible images.
6. **Key:** the private key matches the public key installed at creation, and local permissions are restrictive.
7. **Syntax:** use `ssh -i /path/key user@public-ip`.

Useful SSH diagnostics:

```
ssh -vvv -i ~/keys/oracle-vm.key ubuntu@203.0.113.10
# timeout: usually route/firewall/address
# permission denied (publickey): usually user/key mismatch
```

When an app works locally but not remotely

```
curl -I http://127.0.0.1:PORT
sudo ss -lntup | grep PORT
sudo ufw status verbose
# Then check the OCI NSG/security-list ingress rule for the same port.
```


Treat free infrastructure as replaceable

Cost controls

- ☐ Create a small monthly budget with email alerts.
- ☐ Check Cost Analysis after creating resources.
- ☐ Verify every shape and service is Always Free eligible.
- ☐ Delete unused paid resources, not only their parent VM.
- ☐ Remember: a budget is an alerting tool, not a hard spending cap.

Access controls

- ☐ Keep multifactor authentication enabled for the cloud account.
- ☐ Use SSH keys; keep password login and root login disabled.
- ☐ Restrict SSH ingress to your IP or a private overlay network.
- ☐ Patch the OS and applications regularly.
- ☐ Never store private keys, API keys, or tokens in a public repo.

Budget setup

- 1 Open **Billing & Cost Management › Budgets** and create a budget scoped to the root or project compartment.
- 2 Add actual-spend alert rules at low thresholds appropriate for a free-tier account.
- 3 Send alerts to an address you actively monitor, then verify the alert rule exists.

Idle reclamation changes the design. Oracle may reclaim an Always Free compute instance it classifies as idle. Do not manufacture fake load. Instead, keep important code in source control, back up stateful data, and make the server reproducible so a replacement is routine.

Minimum backup pattern

What	Where	Cadence
Application code	Private Git repository	Every meaningful change
Configuration	Encrypted backup; secrets separate	After configuration changes
Database/data	Encrypted off-instance destination	Based on acceptable data loss
Boot volume	OCI boot-volume backup	Before risky upgrades or migrations

Recovery test: a backup is not proven until you can restore it. Keep a short rebuild note: image, shape, ports, DNS, packages, services, mount points, and restore command.

The one-page memory card

Provisioning words

Tenancy: your whole OCI account.

Region: geographic deployment area.

Compartment: logical resource and policy container.

Shape: VM CPU architecture and resource profile.

Image: operating system template.

OCID: globally unique Oracle resource ID.

Networking words

VCN: virtual cloud network.

Subnet: address range inside a VCN.

VNIC: virtual network interface card.

NSG: firewall rules for selected VNICs.

Security list: firewall rules for a subnet.

CIDR: compact notation for an IP range.

Recommended starter: one Ubuntu LTS Arm VM on VM.Standard.A1.Flex , 2 OCPUs, 12 GB RAM, 100 GB boot volume, reserved public IP, key-based SSH, TCP 22, and public TCP 80/443 for web traffic.

Official Oracle references

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